

Design and Analysis of a Low-Power Conditional-Capture Flip-Flop in 22 nm CMOS Technology

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Abstract—In this paper, the design and implementation of a Low Power Conditional Capture Flip-Flop (CCFF) are presented, employing 22 nm CMOS technology. The main goal is to reduce the dynamic power consumption, especially in the clock network, by avoiding redundant internal flip-flop transitions when the input data is constant, i.e., when the input data 'D' is equal to the output 'Q'. For this purpose, the proposed architecture incorporates data-dependent control logic to disable the internal clocking mechanism, thereby reducing the power consumption. The proposed design is implemented and simulated using the Electric VLSI Design System and LTspice. Simulation results are presented after layout versus schematic (LVS) verification, comparing the proposed CCFF with the conventional Master-Slave Flip-Flop and other standard designs. The proposed CCFF reduces the total power by 42.5%.

I. INTRODUCTION

THE global need for portable electronics, such as tablets, mobile phones, and laptops, has seen an increased demand. For consumers, the key difference between one device and another is often the lifespan of the battery. With consumers preferring devices that can function for an extended period on a single charge, minimizing power consumption has become an important problem for VLSI design.

A key limitation of standard flip-flops lies in their inherent inefficiency. In standard flip-flops, power consumption occurs with every clock pulse, even when there is no change in the data. This behavior is referred to as “redundant switching”, where power is consumed without a corresponding data transition. In addition, research indicates that the clock distribution network alone accounts for approximately 30 - 60% of the total system power [1].

However, reducing power should not come at the cost of performance or excessive silicon area. Critical timing parameters, such as setup time, hold time, and Clock-to-Q delay, must be maintained to ensure the circuit works at high speeds. Furthermore, the physical area of the design directly impacts the manufacturing cost. Therefore, the challenge is to design a flip-flop that saves energy without making the device slower, sensitive to clock skew, or occupying a large footprint.

To overcome this problem, this paper has proposed a more intelligent design for the flip-flop in 22 nm CMOS technology. Consequently, internal nodes may be repeatedly charged and discharged even when the stored data does not change between consecutive clock cycles. This unconditional switching not only wastes power but also increases switching noise and may

adversely affect signal integrity and overall system reliability. Therefore, reducing redundant switching activity inside flip-flops has become a major research focus in low-power VLSI design.

II. LITERATURE REVIEW

Flip-flops are among the most power-critical elements in synchronous VLSI systems because they switch every clock cycle and also contribute significant loading to the clock distribution network [1], [2]. Even when the data input remains constant, conventional edge-triggered flip-flops still exhibit internal charging and discharging activity driven by the clock, leading to redundant dynamic power consumption [2]. This redundancy becomes more dominant in deeply pipelined designs, where an increased number of register stages causes the clock network overhead to represent a large fraction of the total chip power [3].

A. Conventional Master-Slave Flip-Flops

Traditional master-slave (MS) flip-flops guarantee robust edge-triggering by cascading two latching stages [4]. While this structure improves functional correctness and timing isolation, it inherently incurs higher clock-driven switching activity because internal nodes are toggled on every cycle regardless of data activity [5]. As a result, MS flip-flops typically exhibit higher dynamic power consumption under low and moderate switching activity, which is common in many practical workloads [5].

B. Conditional Techniques for Reducing Redundant Switching

To address redundant internal transitions, conditional techniques were introduced to prevent unnecessary charging/discharging when the stored state does not change. In general, these approaches exploit the observation that when $D = Q$, a flip-flop does not need to perform a full internal evaluation to maintain its state; therefore, internal switching can be selectively suppressed.

1) *Conditional-Capture Flip-Flop (CCFF)*: Kong et al. proposed the Conditional-Capture Flip-Flop (CCFF), which reduces internal switching by enabling the capture and evaluation mechanism only when the new input differs from the stored output, i.e., only when a real state change is required. The CCFF achieves statistical power reduction because typical

input sequences contain long runs of identical values, allowing unnecessary transitions inside the flip-flop to be avoided. This design is particularly effective in low-activity regimes and is widely cited as a reference architecture for clock and power reduction in sequential elements [6].

2) *Conditional-Discharge Flip-Flop (CDFF)*: Zhao et al. analyzed low-power flip-flop approaches and classified them into conditional precharge and conditional capture categories, then introduced a third approach based on conditional discharge. In the CDFF design, redundant activity is reduced by controlling the discharge path of internal dynamic nodes, thereby preventing repeated evaluation when the input remains stable, especially in cases where internal nodes would otherwise charge and discharge every clock cycle. Their results also highlight an additional benefit: suppressing redundant internal node transitions reduces output glitches and switching noise, improving signal integrity [7].

These two reference lines, CCFF and CDFF, motivate an important design tradeoff. Conditional capture reduces activity by disabling sampling when no state change is needed, while conditional discharge reduces activity by preventing unnecessary internal evaluations in dynamic stages [7] [8].

C. Motivation and Gap Addressed by This Work

Although prior CCFF and CDFF designs demonstrate substantial savings, their reported results are technology-and-implementation-dependent, and their suitability must be reassessed under modern scaled CMOS conditions and specific layout constraints. Therefore, this work adopts the CCFF principle (data-dependent suppression of redundant internal switching) and implements a low-power CCFF differential design in 22 nm CMOS, emphasizing a clear operating principle tied to the $D = Q$ condition, a consistent pre-layout and post-layout evaluation flow, and a direct comparison against a conventional master-slave flip-flop and the three reference designs across clock power, total dynamic power, timing (setup time/hold time/clock-to-Q, D-to-Q), and area.

III. CONDITIONAL-CAPTURE FLIP-FLOP ARCHITECTURE

A. Structural Overview

The CCFF architecture, as proposed, distinguishes itself from traditional approaches in the incorporation of a transparency window generator and a comparator block. Referring to the schematic representation in Fig. 2, the CCFF architecture is divided into two stages:

- The Differential Sensing Stage: The circuit uses a pair of NOR gates to process the feedback signals from the output nodes Q, Q', and the delayed clock signal CKD.

- The Latching Stage: A cross-coupled inverter pair is utilized to maintain the state of the output node during the opaque period of the clock signal.

B. Operational Principle

The key to the functionality of the circuit is the precharging of the internal nodes. During the Evaluation Phase, when the clock is high, the precharge nodes SB and RB are selectively

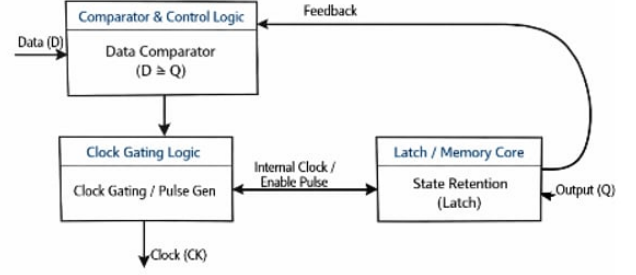


Fig. 1. Block diagram of the proposed Conditional-Capture Flip-Flop illustrating the differential sensing logic, conditional capture mechanism, and latch stage.

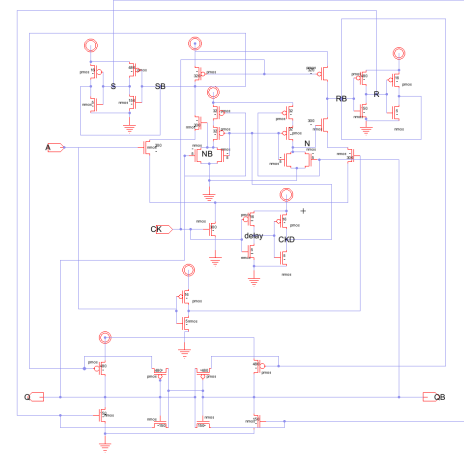


Fig. 2. Gate-level schematic of the proposed Conditional-Capture Flip-Flop in 22 nm CMOS technology.

discharged. The discharge is enabled when the input data D is different from the stored data Q. In such a case, the comparator logic is enabled, allowing the new data to be propagated to the output. In the case where the input data is the same as the stored data, the pull-down path is disabled, causing the precharge nodes to remain precharged. This ensures that there is no dynamic power consumption since the internal nodes remain precharged.

C. Transistor Sizing Strategy

A PMOS/NMOS width ratio of 3.2 ($W_p/W_n = 16/5$) was chosen to compensate for the lower hole mobility in PMOS devices, ensuring symmetrical rise and fall times and centering the switching threshold V_M at $V_{DD}/2$. This specific ratio provides an optimal balance between propagation delay and noise margins without incurring unnecessary area overhead.

Complementary to this, path-specific sizing was optimized based on functional requirements:

- High-Speed Paths: Critical inverter stages utilized width factors of 50 and 100 to enhance drive strength and minimize latency.

- Delay Generation: For paths requiring intentional delay, such as the delayed clock (CKD), minimum transistor widths were combined with increased interconnect lengths (up to 220 nm) to maximize propagation delay while minimizing dynamic power consumption and capacitive loading.

D. Structural Comparison

As opposed to the conventional Master-Slave Flip-Flop that always dissipates dynamic power due to the continuous charging and discharging of the nodes during the clock edges, the proposed CCFF structurally includes a clock gating technique. This technique is able to completely disable the internal clock distribution network when the data is stable ($D = Q$). Furthermore, as opposed to other high-performance reference architectures such as the Hybrid Latch Flip-Flop (HLFF) or the Semi-Dynamic Flip-Flop (SDFF) [9], which often tend to compromise on power to improve speed, the proposed design focuses on the power efficiency of the flip-flop. This is achieved through the differential sensing technique that only becomes active when there is a data mismatch.

IV. PRE-LAYOUT SIMULATION AND ANALYSIS

A. Simulation Environment

The proposed Conditional Capture Flip-Flop design was validated using the LTspice simulation software with the 22 nm High-Performance Predictive Technology Model (PTM). The simulation was carried out at a nominal temperature of 25°C with a supply voltage of 0.8 V and with a Frequency of 600Hz. To drive the circuit with realistic conditions for a digital pipeline, a load capacitance of 400 fF was connected to the output nodes Q and QB.

B. Timing Analysis

The transient analysis was carried out to evaluate the speed of the proposed design. The propagation delay was defined as the time difference between the 50% transition of the clock signal CK and the 50% transition of the output Q during a low-to-high transition. The simulation results show that the proposed design achieves a Clock-to-Q delay of 206.76 ps, thereby confirming the high-speed capability of the differential sensing stage. The rise time of the output Q was also evaluated as 109.34 ps for the 10% to 90% of VDD transition with a heavy load of 400 fF, thereby confirming the robust driving capability of the proposed design.

C. Power Consumption

The power consumption of the proposed design was analyzed by integrating the current supplied by the supply voltage $I(VDD)$ over the 1.8 ns time period of the simulation. The average dynamic power consumption P_{avg} was then calculated using the relation $P_{avg} = V_{DD} \times I_{avg}$. The proposed design achieves an average power dissipation of 179.54 μ W during the active switching phase, as summarized in Table I. This reduction in power consumption is attributed to the conditional capture mechanism, which gates the internal clock to eliminate unnecessary switching activity when the input data remains stable.

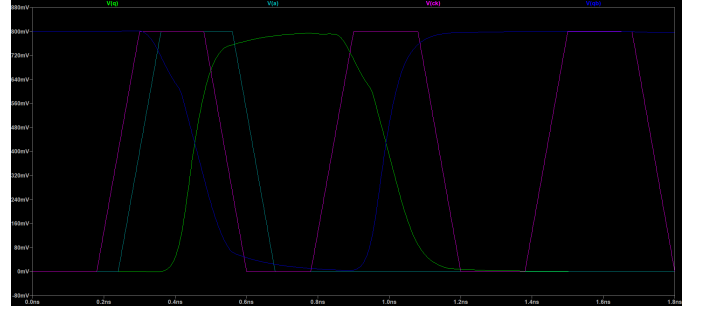


Fig. 3. Pre-layout transient waveforms verifying the correct edge-triggered behavior of the proposed CCFF.

TABLE I
SIMULATION SUMMARY

Parameter	Value	Unit
Technology Node	22 nm HP	
Supply Voltage (VDD)	0.8	V
Temperature	25	°C
Load Capacitance (CL)	400	fF
Clock-to-Q Delay	206.76	Ps
D -to-Q Delay	164	Ps
Rise Time	109.34	Ps
Fall Time	190.61	Ps
Average Power	179.54	μ W

D. Pre-Layout Transient Simulation Waveforms

The waveforms in Fig. 3 confirm correct edge-triggered operation of the proposed CCFF. The output signal Q updates only on the active clock edge and remains stable otherwise, demonstrating proper functional behavior and effective suppression of unnecessary internal switching.

V. LAYOUT DESIGN AND POST-LAYOUT ANALYSIS

A. Physical Layout Implementation

The physical layout of the proposed Conditional-Capture Flip-Flop (CCFF) was implemented using the Electric VLSI Design System. A compact rectangular topology was adopted to optimize both area utilization and performance. The layout follows a standard-cell alignment methodology, in which all PMOS transistors are placed in the upper region and all NMOS transistors are placed in the lower region. This organization enables efficient N-well sharing and minimizes isolation overhead, resulting in a dense and manufacturable layout.

B. Routing Strategy and Design Rule Compliance

Routing was performed using a hierarchical metal stack spanning metal layers M1 through M6. Local interconnections were primarily routed using the lower metal layers (M1-M3), while higher metal layers (M4-M6) were reserved for long-distance routing and global clock distribution. This routing strategy reduces parasitic resistance and capacitance along critical signal paths and improves overall signal integrity.

All layout geometries were designed in full compliance with the 22-nm high-performance CMOS design rules, including minimum metal width, spacing, and enclosure constraints. Design-rule checking (DRC) was performed using Electric to ensure strict adherence to all technology constraints.

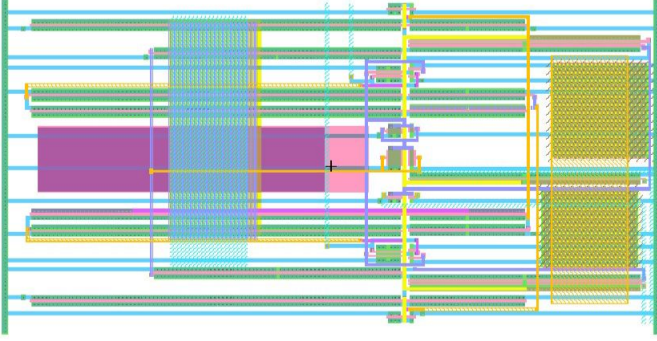


Fig. 4. Physical layout of the proposed Conditional-Capture Flip-Flop implemented in 22-nm CMOS technology.

C. Parasitic Extraction and Layout Area Characterization

Layout-versus-schematic (LVS) verification was conducted to ensure structural consistency between the schematic and the physical layout. Following successful LVS verification, an RC-extracted SPICE netlist was generated to accurately model the lumped parasitic resistance and capacitance associated with interconnects across metal layers M1-M6. This extracted netlist was subsequently used for post-layout simulations to evaluate the impact of parasitic effects on circuit performance.

The final layout occupies a total cell area of $47.02 \mu\text{m}^2$, with a total interconnect length of approximately $380.72 \mu\text{m}$. At the transistor level, the proposed CCFF exhibits a total gate width of $73.007 \mu\text{m}$, reflecting a compact and efficient implementation. Due to the narrow metal widths characteristic of the 22-nm technology node, interconnects contribute only a small fraction of the total cell area, while the majority of the layout area is dominated by transistor active regions.

D. Comparison of Pre- and Post-Layout Performance

To assess the impact of parasitic extraction on circuit behavior, a detailed comparison between pre-layout and post-layout simulation results was performed. The results are summarized in Table II.

The post-layout clock-to-Q delay Decreased by only 1.75%, indicating minimal sensitivity to parasitic effects. Similarly, the observed increase in average power and current remained below 4%, confirming that the conditional-capture mechanism effectively suppresses redundant switching even after layout parasitics are introduced. The rise-time degradation is limited to 2.96%, while the energy per operation increases by 1.27%, remaining within acceptable margins for low-power sequential elements.

Overall, These results confirm that the physical implementation preserves the low-power and high-speed characteristics observed at the schematic level.

E. Performance Degradation and Margin Analysis

To evaluate the robustness of the proposed CCFF, a sensitivity analysis across temperature and voltage variations was performed. As shown in Table III, the design maintains functional integrity and a stable output (Q) with negligible

TABLE II
PRE-LAYOUT VS. POST-LAYOUT PERFORMANCE COMPARISON

Metric	Pre-Layout	Post-Layout	Change
Clock-to-Q Delay (ps)	206.76	203.18	1.75%
D-to-Q Delay (ps)	164.67	162.67	2.00%
Average Power (μW)	179.54	185.05	3.02%
Average Current (μA)	224.42	231.32	3.03%
Rise Time (ps)	109.34	112.63	2.96%
Fall Time (ps)	190.77	193.83	1.59%
Energy (fJ)	37.12	37.59	1.27%
Avg. Clock Power (μW)	52.65	57.05	8.57%

fluctuations. Although a performance inflection occurs at 0.7 V as a result of the diminished driving current, the CCFF remains fully operational.

The performed tests under an aggressive 400 fF load demonstrates high-fanout reliability. Given that delay scales linearly with load, the design achieves significantly improved power and timing in typical low-load scenarios.

VI. RESULTS AND COMPARISON

Table IV presents the post-layout performance of the proposed Conditional-Capture Flip-Flop (CCFF) in comparison with representative flip-flop designs reported in the literature. The proposed design was evaluated using a load capacitance of 400 fF and a data activity factor of one-third, matching the operating conditions used in Reference 6 to ensure a fair comparison.

Under identical operating conditions, the proposed CCFF demonstrates a clear reduction in dynamic power compared to the original conditional-capture flip-flop, primarily due to improved suppression of redundant internal switching. In contrast, References 7 and 8 report their results under a significantly smaller load capacitance of 20 fF, which results in a different comparison basis and explains the variation in the reported power values.

In terms of timing performance, the proposed CCFF maintains competitive clock-to-Q delay with adequate setup and hold margins, while exhibiting minimal degradation after layout parasitic extraction. Overall, the results confirm that the proposed design achieves an effective balance between power efficiency and timing performance under realistic operating conditions.

VII. FUNCTIONAL VERIFICATION

To demonstrate practical operation, a 32-bit register bank was implemented using the proposed CCFF and simulated under realistic input activity. The waveform in Fig. 5 shows that the register correctly captures the input data on the active clock edge and preserves the stored value between clock transitions. As a result, the multi-bit register operates reliably while providing an optimal balance of high-speed data path performance, low dynamic power consumption, and a compact implementation area.

TABLE III
PERFORMANCE SENSITIVITY TO TEMPERATURE AND VOLTAGE VARIATIONS

Condition	Clock-to-Q (ps)	Rise Time (ps)	Fall Time (ps)	Total Power (μ W)	Clock Power (μ W)	PAP (μ W $\cdot$ μ m ²)
Temperature Variation (0.8V)						
0°C	185.71	88.07	171.08	183.29	58.18	8618.29
25°C	203.18	112.63	193.83	185.05	57.05	8701.05
60°C	231.17	193.62	282.63	185.29	56.73	8712.33
Voltage Variation (25°C)						
0.7V	256.83	210.57	369.73	137.29	44.05	6455.37
0.8V	203.18	112.63	193.83	185.05	57.05	8701.05
1.0V	149.22	93.31	145.25	322.89	87.52	15182.28
1.2V	115.13	90.19	125.45	660.3	119.97	31047.30

TABLE IV
POST-LAYOUT PERFORMANCE COMPARISON OF THE PROPOSED CCFF WITH PUBLISHED FLIP-FLOP DESIGNS

Design	Clock Power (μ W)	Total Dynamic Power (μ W)	Area / Gate Width	Setup Time (ps)	Hold Time (ps)	D-to-Q Delay (ps)
Proposed CCFF	57.05	185.05	47.02 μ m ² / 73.007 μ m	-40	50	162.67
Kong et al. CCFF [6]	Not reported	322	Not reported / 146 μ m	-50	200	337
Zhao and Roy CDFF [7]	Not reported	20.2	Not reported	Not reported	Not reported	185
Roy and Prasad PT-FF [8]	8.03	18.02	79.17 μ m ² / Not reported	-39.7	85.1	107.24

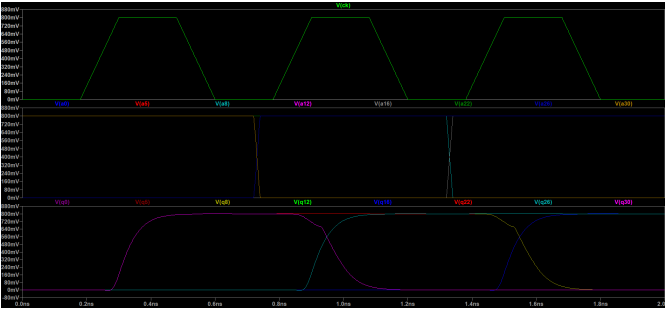


Fig. 5. Transient simulation waveforms of the 32-bit register bank using the proposed CCFF.

VIII. CONCLUSION AND FUTURE WORK

In this paper, a low-power Conditional-Capture Flip-Flop implemented in 22 nm CMOS technology was presented. The proposed design reduces dynamic power consumption by conditionally disabling internal switching activity when the input data remains unchanged. Through a differential sensing mechanism and data-dependent clock gating, redundant transitions inside the flip-flop are effectively eliminated.

Pre-layout and post-layout simulation results demonstrate that the proposed CCFF maintains competitive timing performance while achieving significant power savings, with minimal degradation after parasitic extraction. The layout results further confirm the suitability of the proposed architecture for scaled CMOS technologies.

Future work will focus on extending the proposed CCFF design to multi-bit register banks in order to evaluate system-level power savings under different data activity patterns. Additional optimizations, such as adaptive clock gating and further technology scaling, can also be explored to improve energy efficiency in large-scale digital systems.

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